

**DESCRIPTION**

ALZ 3721 is a precision submersible hydrostatic level transmitter. High-quality silicon piezoresistive sensor, electronic module with an active temperature compensation technology, robust stainless steel housing and a hydrometric cable with the ventilation tube make ALZ 3721 an outstanding choice for the applications where better accuracy is required. An optional integrated Pt100 temperature sensor can be supplied for combined level and temperature measurement.

SPECIFICATIONS

Level: 1 mH₂O up to 250 mH₂O

Basic accuracy: $\pm 0.1\%$

Probe: Ø27 mm, AISI 316L stainless steel

Outputs: 4...20 mA; 0...20 mA; 0,5...4,5 B; RS-485 / Modbus RTU; HART®

Sensor: silicon piezoresistive

Media temperature: -20...+70 °C

HART® is a registered trademark of the HART Communication Foundation.

APPLICATIONS

Water, wastewater

Fuels, oils

Tanks, wells

Pump protection

Laboratory equipment

TECHNICAL SPECIFICATIONS

MEASURING RANGES

Pressure range, bar	Level, mH ₂ O	Overpressure, bar	Pressure range, bar	Level, mH ₂ O	Overpressure, bar
0...0.1	0...1	1	0...2.5	0...25	6
0...0.16	0...1.6	1	0...4	0...40	15
0...0.25	0...2.5	1	0...6	0...60	15
0...0.4	0...4	1	0...10	0...100	30
0...0.6	0...6	3	0...16	0...160	60
0...1	0...10	3	0...25	0...250	60
0...1.6	0...16	6	-	-	-

PERFORMANCE

	P > 0.1 bar	P ≤ 0.1 bar
Accuracy, % of span*	≤ ±0.1	≤ ±0.2
Temperature effect (% of span / 10 °C)	≤ ±0.02	≤ ±0.04
Compensated range	-20...+70 °C	0...+70 °C
Power supply effect	≤ ±0.05% of span / 10 V	
Load resistance effect	≤ ±0.05% of span / kOhm (transmitters with current output)	
Long-term stability	≤ ±0.1% of span / year	
Response time (10...90%)	≤ 1 ms with analog output	

* Accuracy includes non-linearity, hysteresis and non-repeatability.

OPERATING CONDITIONS

Medium temperature (depends on seal)	-20...+70 °C		
Cable service temperature (static applications)	PVC: -20...+70 °C PUR: -50...+70 °C		
Storage temperature	-20...+70 °C		
Approval	0Ex ia IIC T6...T4 Ga X		
Temperature class	T4	T5	T6
Ambient temperature	-20...+70 °C	-20...+60 °C	-20...+50 °C
Vibration resistance	10 g RMS, 25–2000 Hz		
Shock resistance	100 g / 11 ms		
Service life	> 100 x 10 ⁶ cycles		

MECHANICAL SPECIFICATIONS

Housing	stainless steel 316L (1.4404)
Seal	FKM; EPDM; NBR
Diaphragm	stainless steel 316L (1.4435)
Cable external jacket	PVC, PUR, FEP
Ingress protection	IP69K

ELECTRICAL SPECIFICATIONS

Output signal	Power supply, U _s	Load resistance, R	Power consumption
4...20 mA / 2-wire	12...36 V	≤ [(U _s – 12 V) / 0.02 A] Ohm*	≤ 26 mA
4...20 mA / HART®			
4...20 mA / 3-wire	5 V	≤ 500 Ohm	≤ 2 mA
0.5...4.5 V / 3-wire		≥ 5kOhm	
0.5...4.5 V / 3-wire	6...15 V		≤ 7 mA
RS-485 / Modbus RTU	12...36 V	-	

* For output signal 4...20 mA / HART® minimum load resistance for digital communication: 250 Ohm.

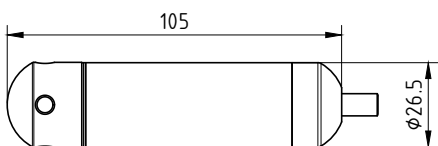
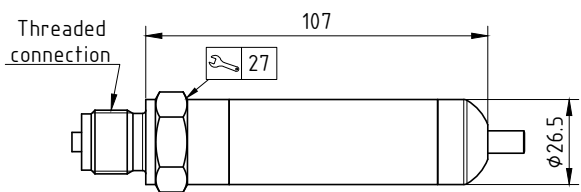
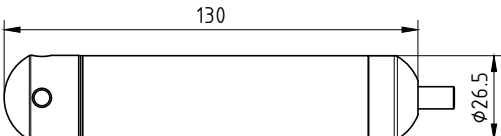
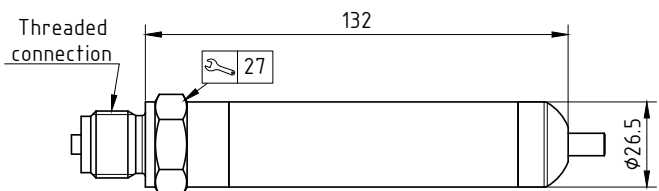
Safe values for intrinsically safe design 0Ex ia IIC T6...T4 Ga X:

Parameter	2-wire	3-wire
Maximum voltage, U _i	28 V	6 V
Maximum current, I _i	93 mA	60 mA
Maximum power, P _i	660 mW	100 mW
Maximum internal inductance, L _i	10 µH	10 µH
Maximum internal capacitance, C _i	15 nF	500 nF

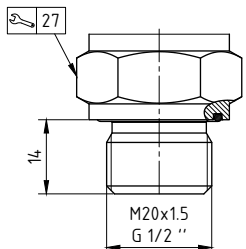
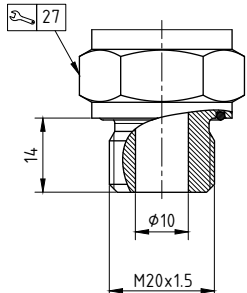
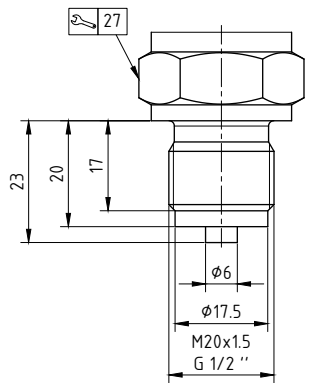
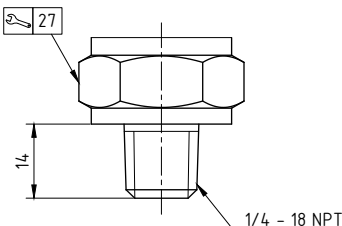
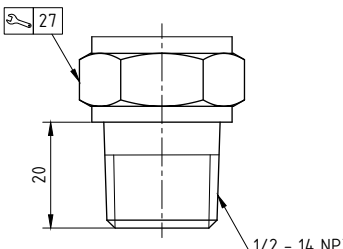
ELECTRICAL CONNECTIONS / PIN ASSIGNMENT

Circuits		Cable gland wires
2-wire	power +	white (red)
	power -	brown (blue)
	ground	yellow-green
3-wire	power +	white (red)
	power -	brown (blue)
	signal +	green (black)
	ground	yellow-green
Pt100 (optional)	T+	yellow
	T-	pink (white)
	T-	gray (green)
RS-485	power +	white (red)
	power -	brown (blue)
	A	yellow
	B	green (black)
	ground	yellow-green
HART®	power +	white
	power -	brown
	ground	yellow-green

DIMENSIONS (mm)

Standard version	Threaded version G1/2 EN; Code 73
	
with HART or Ex ia	with HART or Ex ia
	

THREADED CONNECTIONS (NIPPLES), DIMENSIONS (mm)

M20x1.5 DIN 3852; Code 20 G1/2" DIN 3852; Code 72	M20x1.5 DIN 3852 open port; Code 26	M20x1.5 EN 837; Code 21 G1/2" EN 837; Code 73
		
1/4" NPT; Code 74	1/2" NPT; Code 82	
		

ORDERING CODE

ALZ 3721		-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
UNIT OF MEASUREMENT									
	bar	B							
	kg/cm ²	S							
	mH ₂ O	W							
	kPa	K							
Other (specify when ordering)		X							
UPPER RANGE LIMIT (URL)									
bar, kg/cm ²		mH ₂ O		kPa					
0.1	0100	1	1000	10	1001				
0.16	0160	1.6	1600	16	1601				
0.25	0250	2.5	2500	25	2501				
0.4	0400	4	4000	40	4001				
0.6	0600	6	6000	60	6001				
1	1000	10	1001	100	1002				
1.6	1600	16	1601	160	1602				
2.5	2500	25	2501	250	2502				
4	4000	40	4001	400	4002				
6	6000	60	6001	600	6002				
10	1001	100	1002	1000	1003				
16	1601	160	1602	Other	XXXX				
25	2501	250	2502						
Other	XXXX	Other	XXXX						
ACCURACY									
0.1% (P > 0.1 bar) (standard)				A					
0.2% (P ≤ 0.1 bar) (standard)				B					
Other (specify when ordering)				X					
OUTPUT SIGNAL									
4...20 mA / 2-wire (standard)				A					
4...20 mA / 2-wire, 0Ex ia IIC T6...T4 Ga X				Q					
4...20 mA / 3-wire				B					
0.5...4.5 V / 3-wire, U _s = 5 V, 0Ex ia IIC T6...T4 Ga X				R					
0.5...4.5 V / 3-wire, U _s = 6...15 V				K					
RS-485 / Modbus RTU				M					
4...20 mA / HART®				H					
Other (specify when ordering)				X					
SEAL									
FKM				F					
NBR				N					
EPDM				E					
Other (specify when ordering)				X					
VERSION									
Standard				00					
Pt100 temperature sensor				0T					
Threaded version, M20x1.5 DIN 3852				20					
Threaded version, M20x1.5 EN 837				21					
Threaded version, M20x1.5, open port				26					
Threaded version, G1/2" DIN 3852				72					
Threaded version, G1/2" EN 837				73					
Threaded version, 1/4" NPT				74					
Threaded version, 1/2" NPT				82					
Threaded version G1/2 EN				NC					
Other (specify when ordering)				XX					

ORDERING CODE (CONTINUED)

ALZ 3721	-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
CABLE EXTERNAL JACKET								
							PVC	P
							PUR	U
							FEP	T
							Other (specify when ordering)	X
CABLE LENGTH								
Any length in meters (e.g., 010M means 10 meters)								XXXM

Example: ALZ 3721-W-6000-A-A-F-00 // P-010M

ACCESSORIES

				
Anchoring clamp for submersible transmitter	PZ 1024 Power supply unit	BZ 05 / BZ 10 Dry air junction box for submersible transmitters		